

TOTO Ltd

5332 JP — The toilet maker with a semiconductor engine

RATING: BUY

12-MO TARGET: ¥7,200 (+18%)

Price (2 Oct 2026)	¥6,097
Market cap	~¥1.0tn / US\$6.7bn
EV (e)	~¥937bn
52-week range	¥3,766 – ¥9,500
P/E FY3/27E / FY3/28E	21.8x / 17.3x
EV/EBITDA FY3/26A	10.6x
Dividend yield	2.0%
Ceramics share of FY3/26 segment OP	~51%
Probability-weighted value	~¥6,975 (+14%)

Contents: investment summary; layman's guide; price catalysts; business model; macro; moats; supply chain; 5-year financials & KPIs; earnings & sentiment; EPS forecasts; capital structure; valuation & regression; M&A; management; board; shareholders & activism; strategy; stakes; capital returns; adversarial analysis; scenarios; questions for management; short-seller view; catalysts.

Prepared for buy-side investment committee review. Figures marked (e)/~ are estimates — see Data Notes (Section 25).

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1. Investment Summary

Rating: BUY (moderate conviction, event-driven sizing). 12-month target ¥7,200 (+18% vs ¥6,097 close on 2 Oct 2026); probability-weighted value ~¥6,975 (+14%). TOTO is no longer a toilet company with a ceramics hobby: in FY3/26 Advanced Ceramics (electrostatic chucks for NAND cryogenic etch) generated ¥28.9bn of operating profit on ¥67.4bn of sales (42.9% margin), exceeding the entire ¥670bn housing business for the first time. The stock has round-tripped from ¥9,500 (23 Jun 2026) to ¥6,097 on a Q1 miss, giving a second entry point into a business mix shift that the consolidated P&L still obscures.

Market data (2 Oct 2026)	Value	Valuation	FY3/26A	FY3/27E	FY3/28E
Share price	¥6,097	P/E	25.1x	21.8x	17.3x
52-week range	¥3,766 – ¥9,500	EV/Sales	1.27x	1.22x	1.17x
Shares outstanding (ex-treasury)	~164.4m	EV/EBITDA	10.6x	9.8x	8.1x
Market cap	~¥1.00tn (~US\$6.7bn)	EV/EBIT	17.4x	15.6x	12.0x
Net cash (e)	~¥65bn	Dividend yield	1.8%	2.0%	2.1%
Enterprise value (e)	~¥937bn	Net debt/EBITDA	-0.7x	-0.7x	-0.7x
Consensus target (8 analysts)	~¥7,785–7,955	ROE	7.7%	~8.6%	~10.3%

Sources: company filings (FY3/26 results 30 Apr 2026; Q1 FY3/27 31 Jul 2026), Kabuyoho/Minkabu/TipRanks consensus; FY3/27E–FY3/28E are our estimates. (e) = estimate, balance sheet not directly verified — see Data Notes.

Thesis in five points

- **Hidden compounder inside a mature franchise.** Ceramics OP has grown from ~¥9bn (FY3/22e) to ¥28.9bn (FY3/26) and is guided to ~58% of group OP in FY3/27. Replacement demand (~80% of segment sales per Palliser) and Lam Research's cryo-etch installed base (expected to triple in 3–5 years) underpin a multi-year growth runway; capacity +20% from Jan 2027 (Buzen) and a ~¥30bn (FY28) / reported ~¥80bn (5-year) investment programme.
- **Housing at a cyclical and structural trough, with self-help.** Japan new-build starts fell 12.9% in FY2025 to ~711k; China lost ¥6.9bn in FY3/26. But two China plants are closed (~¥7bn profit improvement targeted for FY3/27), US capacity +150% (Georgia), and an 8–13% Japan price increase lands 1 Dec 2026 — the largest in a decade.
- **Activist has already won the first round.** Palliser Capital (Feb 2026) asked for ceramics disclosure, more ceramics investment, housing/China restructuring and capital efficiency. TOTO adopted the substance on 30 Apr 2026 (Palliser statement, 7 May 2026). Round two — cross-shareholding sell-down, excess cash, balance-sheet efficiency (equity ratio 63.8%) — is the likely content of the next mid-term plan (STAGE3, ~Apr 2027).
- **Valuation reset.** At ¥6,097 the market implies ~¥615bn for ceramics (~19x FY3/27E EBIT, ~15x FY3/28E) if housing is valued on the peer margin regression — not demanding for a ~43%-margin, sole/dual-source semiconductor consumable versus NGK Insulators at 22x forward P/E.
- **The bear case is real and cyclical, not existential.** FY3/24 showed ceramics OP can fall ~35% in a memory capex downturn; customer concentration (Lam) is high; Q1 FY3/27 showed shipment lumpiness. Our bear value of ~¥3,500 reflects a full memory downturn plus housing de-rating.

What has to go right / what would make us wrong

Must go right (next 2–3 quarters)	Would change our view
H1 FY3/27 (~30 Oct 2026) shows Q1 ESC shipment slip was timing, ceramics H1 OP ≥ ¥15bn	Ceramics H1 OP down y/y with no customer-timing explanation; guidance for ceramics sales (¥85.5bn) cut
China returns to near break-even as plant closures flow through	China losses persist beyond FY3/27, implying further impairment
Dec 2026 Japan price hike sticks (TOTO ~60% toilet share; LIXIL follows)	Volume loss > price; LIXIL/Panasonic do not follow
STAGE3 plan (Apr 2027) adds capital-allocation targets: ROE ≥10–12%, cross-holding exit, buybacks	Cash hoarded, no ROE/cross-holding targets; Palliser disengages
NAND WFE continues to recover (GS: \$11bn/15bn/22bn for 2026/27/28)	Memory capex cut; TEL cryo-etch wins displace Lam/TOTO

2. Layman's Guide: What TOTO Does and How It Makes Money

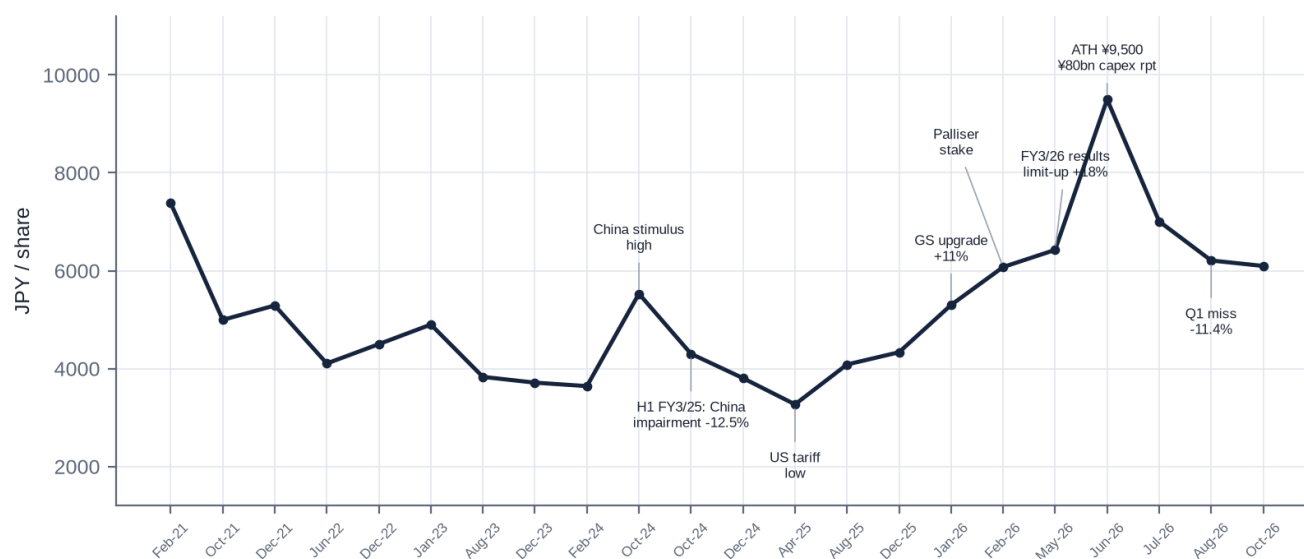
In one sentence: TOTO makes the toilets, heated "Washlet" bidet seats, taps, bathtubs and vanity units that sit in roughly six out of ten Japanese bathrooms — and, in a small but very profitable side business, makes ultra-precise ceramic "chucks" that hold silicon wafers in place inside the machines that etch memory chips.

Division	What it sells	Who pays	FY3/26 sales	FY3/26 OP	Share of OP
Japan Housing Equipment	Toilets, Washlet, unit baths, kitchens, vanities, faucets; ~72% is remodelling/replacement	Plumbing wholesalers → contractors, homebuilders, remodelers, homeowners	¥479.7bn (65%)	¥20.3bn (4.2%)	~36%
Overseas Housing – China	Premium toilets/Washlet in mainland China	Distributors, developers	¥53.8bn (7%)	~¥6.9bn	loss
Overseas Housing – Americas	Washlet, high-end toilets (NEOREST); plants in Georgia (US), Mexico	Plumbing distributors, showrooms, retail	¥75.6bn (10%)	~¥4.7bn	~8%
Overseas Housing – Asia/Oceania	Sanitary ware, faucets; Vietnam, Thailand, India plants	Distributors, hotels, developers	¥54.9bn (7%)	¥10.2bn	~18%
Advanced Ceramics (New Domain)	Electrostatic chucks (ESCs) for 3D NAND cryo-etch; aerosol-deposition chamber parts; LCD parts	Semiconductor equipment makers — principally Lam Research	¥67.4bn (9%)	¥28.9bn (42.9%)	~51%
Group			¥737.4bn	¥53.8bn (7.3%)	100%

Segment OP before ~¥3bn of corporate costs/eliminations; share computed on segment total ¥56.8bn. China/Americas/Asia from FY3/26 results commentary; "Other overseas" (~¥5.7bn sales) not shown.

- **How the toilet business makes money:** a mature, oligopolistic replacement market. Japanese toilets are replaced every ~15–20 years and Washlet seats every ~10; TOTO earns a premium price for brand and reliability, sells through wholesalers on standard trade credit, and supports demand via ~100 showrooms and a network of "Remodel Club" contractors. Margins are thin (~4%) because of high fixed costs, rising materials/labour and a shrinking new-build market.
- **How the ceramics business makes money:** an electrostatic chuck is the "table" inside an etch chamber that grips the wafer with static electricity and controls its temperature to within fractions of a degree, at –60°C to –100°C in cryogenic NAND etch. It wears out and is replaced roughly annually, so most revenue is recurring. TOTO has co-developed chucks with Lam Research since ~1990 and is reported to be the #2 global ESC maker; margins are >40%.
- **Why it matters:** 9% of sales now produce over half the profit — and the market values those two businesses very differently.

3. Historical Share-Price Catalysts (Moves >10%, 2021–2026)



TOTO share price, key levels Feb 2021 – Oct 2026 (JPY). Reconstructed from annual OHLC and reported event closes; intermediate points are illustrative.

Date	Move	Catalyst	Read-across
Feb–Oct 2021	¥7,380 → ¥4,995 (-32%)	Post-COVID housing peak unwinds; Evergrande/China property contagion fears (Nikkei: "TOTO plunges on China risk")	China was ~15% of sales and the growth engine — the start of a 4-year de-rating
Jan–Jun 2022	¥5,380 → ¥4,105 (-24%)	Material/energy inflation, China lockdowns, global growth-to-value rotation	Margin squeeze; FY3/23 OP fell 6%
10 Jan 2023	Up to +10% intraday	China reopening hopes	Short-lived
Jul–Aug 2023	Slide to ¥3,831 (3-yr low)	China property slump; weak China sales	—
30–31 Oct 2023	Sharp fall	FY3/24 OP guidance cut to ¥47bn; China OP guided -30% to ¥5.7bn (half plan)	First explicit China earnings reset
Feb–Oct 2024	¥3,642 → ¥5,530 (+52%)	China stimulus rally; early ceramics recovery	Proved China-beta still dominated the stock

Date	Move	Catalyst	Read-across
29 Oct 2024	-12.5% (-¥614)	H1 FY3/25: China swung to a ¥4.4bn operating loss; ~¥34bn China impairment; NI guidance cut	Capitulation on China
2–9 Apr 2025	To ¥3,269 (5-yr low)	US "Liberation Day" tariffs (Vietnam, Mexico, Thailand, Malaysia sourcing)	Low point of the cycle
28 Apr 2025	Positive	FY3/25 results; ¥20bn (4.7%) buyback, later cancelled; China plant closures	First large buyback in years
25 Aug 2025	+8.4%	Americas plan: ¥100bn sales by FY3/31 (+40%); US local production >50%	—
Nov 2025 – Jan 2026	Steady re-rating	H1 FY3/26 ceramics OP ~¥12.9bn; AI/NAND narrative	Market starts to price ceramics
22 Jan 2026	+11% intraday (+8.8% close)	Goldman Sachs upgrade Neutral → Buy on NAND ESC thesis	Largest daily gain since Feb 2021 at the time
17 Feb 2026	+5% day; ~+40% YTD by late Feb	Palliser Capital discloses stake; "Value Enhancement Plan" (>55% upside, ~¥8,800 intrinsic)	Activism catalyses SOTP framing
1 May 2026	Limit-up, ~+18% to ¥6,425	Record FY3/26; ceramics disclosure; ¥30bn ceramics capex; DPS ¥110 → ¥120 guide; buyback	Record single-day gain
19–23 Jun 2026	~+9–11%; ATH ¥9,500	Nikkei: ~¥80bn 5-year chip-parts investment incl. 1nm-era R&D	Peak "AI beneficiary" multiple
3 Aug 2026	-11.4% to ¥6,209	Q1 FY3/27 miss (sales ¥168.6bn vs ~¥176.5bn cons.; EPS ¥42 vs ¥45.6); ESC shipment timing; Middle East (-¥3bn OP); yen strength	Momentum broken; broker target cuts (MS ¥5,900, Nomura ¥6,820)

Sources: Nikkei price history, Nikkei/Bloomberg/Japan Times/Kabutan/Zaikei reporting. Exact daily moves not all verifiable from primary exchange data.

4. Business Model, Key Drivers, Customers and Competitors

4.1 Key drivers

Driver	Housing equipment	Advanced Ceramics
Volume	Japan remodel (~¥347bn, ~72% of Japan sales) driven by ageing stock (~15–20yr replacement); new-build (¥133bn) tied to starts (711k in FY2025, -12.9%); overseas: US Washlet adoption, Asia urbanisation, China property	3D NAND layer count (→ ~1,000 layers by 2030), cryo-etch tool installations (Lam), ESC replacement cycle (~1 year), logic etch AD-coated parts
Pricing	Annual Japan list-price rises: Aug-23 (3–8%), Aug-24 (2–11%), Oct-25 (2–5%), Dec-26 (8–13%, Washlet ~10%). US ~+10% Jun-2025 (tariffs)	Engineered, qualified-in parts; price set per tool generation; strong pricing given ~5-yr technology lead claimed by Palliser
Mix	Washlet/NEOREST premium mix, remodel vs new-build, overseas high-end	Cryo-NAND ESC (highest margin) vs LCD parts; new AD parts for logic
Costs	Brass/copper, resin, energy (kilns), logistics, wages (Japan labour shortage); tariffs on US imports	Alumina/AIN powders, yield, depreciation from new kilns
Capex cycle	Strategic capex ¥175bn under STAGE2 (FY24–26): global ¥72bn, ceramics ¥29bn, domestic ¥32bn, IT ¥42bn	WFE cycle: SEMI 300mm fab equipment +18% (2026) / +14% (2027); GS NAND WFE \$11bn/15bn/22bn (2026–28)
Regulation	Japan building-code/energy rules (Apr 2025) depressed starts; water-efficiency standards favour premium fixtures; US tariffs	Export controls (China fabs), FEFTA designation of semiconductor parts as core sector

4.2 Customers, suppliers, competitors, geographies

- **Customers (housing):** plumbing/building-material wholesalers (e.g. Hashimoto Sogyo, Watanabe Pipe — channel partners, TOTO-specific volumes not disclosed), homebuilders (Sekisui House, Daiwa House, Sumitomo Forestry, Iida Group), condo developers, remodel contractors (TOTO Remodel Club, TDY alliance with Daiken and YKK AP), home centres; overseas: US plumbing distributors/showrooms (e.g. Ferguson) and e-commerce, Asian distributors and hotel projects.
- **Customers (ceramics):** Lam Research (primary; co-development since ~1990; Supplier Excellence Award 2023 and 2024). Applied Materials and Tokyo Electron are potential/partial customers (not confirmed). End users: Samsung, SK hynix, Kioxia/SanDisk, Micron, YMTC (subject to export rules).
- **Suppliers:** ceramic raw materials (kaolin/ball clay, feldspar, silica), brass and copper alloys (faucets), resins (seats/lids), electronic components and PCBs (Washlet), packaging; for advanced ceramics, high-purity alumina and aluminium nitride powders (Tokuyama is the global AIN leader — TOTO's supplier list is not disclosed).
- **Competitors:** Japan — LIXIL (INAX; #2 in toilets), Panasonic Housing Solutions (Alauno), Takara Standard and Cleanup (kitchens/baths). Global — Kohler (private), Roca (private, owns Laufen), Geberit (Duravit is private), Masco (Hansgrohe/Delta), LIXIL (American Standard, Grohe), Villeroy & Boch (+Ideal Standard), Jomoo, Arrow Home, Huida, Hegii (China). ESC — Applied Materials and Lam in-house, Shinko Electric (JIC-owned since 2025), NGK Insulators, Kyocera, Nitterra/NTK Ceratec, Entegris, Sumitomo Osaka Cement, MiCo (Korea), CoorsTek (private).
- **Geographies:** Japan ~65% of sales; Americas ~10%; China ~7% (from ~15% in FY3/22); Asia/Oceania ~7%; ceramics (shipped globally, mostly to US/Asian tool makers) ~9%.

- **Contracts and payment terms:** not disclosed in detail. Housing sales are spot/annual-price-list sales to wholesalers on Japanese trade-credit terms (typically 60–120 days incl. notes); ceramics sold under qualified-supplier agreements with OEMs; revenue recognised on shipment/delivery (relevant to the Q1 FY3/27 "timing" miss).

5. Macro and Micro Factors: Tailwinds and Headwinds

Tailwinds	Headwinds
AI-driven NAND upgrade cycle; cryo-etch tool base to triple in 3–5 yrs (Lam); WFE "low-\$150bn" in 2026	Memory capex cyclicality — FY3/24 ceramics OP fell ~35% (e); single-customer concentration
ESC replacement demand (~80% of segment revenue recurring per Palliser)	TEL entering cryogenic etch could shift chuck sourcing
Japan remodel market ~¥8–9tn, ageing housing stock; Washlet replacement cycle	Japan housing starts at post-war lows (711k FY2025; NRI ~610k by 2040)
Price increases (Dec 2026: 8–13%) in a ~60%-share oligopoly	Labour and materials inflation in Japan; Q1 FY3/27 Japan OP -46%
China restructuring: ~40% capacity removed, ~¥7bn improvement targeted	China property deflation; local competitors (Jomoo, Arrow, Hegii) gaining at lower price points
US Washlet adoption, Georgia plant (+150% capacity), Americas ¥100bn target	US tariffs on Vietnam (20%), Mexico, Thailand/Malaysia (Washlet sourcing)
Weak yen supports overseas profit translation and ceramics exports	Yen strength after intervention (Aug 2026); Middle East disruption (-¥3bn Q1 OP)
Corporate-governance reform in Japan (TSE P/B push, cross-holding unwind, activism)	Large stable-shareholder base slows change

6. Competitive Position and Economic Moats

TOTO is the Japanese category-defining brand ("Washlet" is a TOTO trademark that became the generic term). In Japan it holds ~60% of toilets against LIXIL; Japanese household penetration of warm-water bidet seats exceeds 80%. Overseas it is a premium niche brand with high awareness among travellers to Japan but small share against Kohler, American Standard and local champions. In ceramics, TOTO is reported as the #2 global ESC maker and a primary supplier for Lam's cryogenic dielectric etch tools.

Moat category	Housing equipment	Advanced Ceramics	Strength
Brand	Very strong in Japan (Washlet generic); premium perception in US/Asia	B2B — reputation for quality/yield with Lam	Strong
Hard assets	~20 plants worldwide; kilns and casting know-how; logistics/showroom network	Proprietary sintering/bonding; new Buzen kiln, Nakatsu, Chigasaki R&D	Moderate–Strong
Long-term contracts	None (price-list sales)	Qualified-supplier positions — designed into tool platforms for a generation	Moderate
Network effect	Remodel Club contractor network, showroom footfall	Co-development loop with Lam	Weak–Moderate
Regulatory / switching costs	Low for end user; moderate for builders (spec standardisation)	High: requalification of a chuck takes 12–24 months; process-of-record risk for fabs	Strong (ceramics)
Oligopoly / monopoly	Japan duopoly (TOTO ~60%, LIXIL), Panasonic third	Top-4 ESC suppliers ~90% of market	Strong
Cash-flow predictability	High (replacement-led), but low margin	Medium — recurring replacement but WFE cyclical	Moderate
Scalability	Low — capital-intensive, mature	Medium — capacity-constrained; each +20% requires a new kiln	Moderate
Government interference risk	Low in Japan; tariffs in US; China policy	Export controls; FEFTA protects from foreign takeover	Moderate
Bargaining power	Strong vs wholesalers; weak vs large homebuilders; price leadership in Japan	Balanced — Lam is concentrated buyer but TOTO is hard to replace	Moderate–Strong

Our assessment.

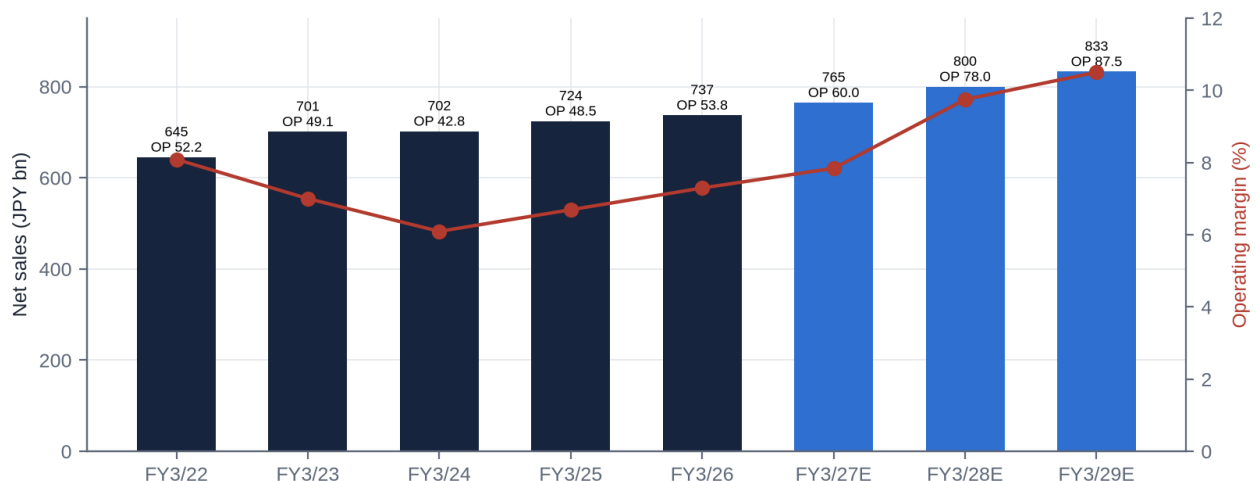
Product comparison. TOTO's Washlet/NEOREST line carries a 20–40% price premium over LIXIL/INAX and Panasonic equivalents in Japan, justified by reliability, self-cleaning glaze (CeFIONtect), Tornado flush and service network. In the US, TOTO is positioned above Kohler/American Standard in bidet seats but with smaller distribution; Kohler out-spends it on marketing. In China, local brands now match features at 30–50% lower prices, eroding TOTO's premium — the root cause of the China impairment.

7. Supply Chain Map

Stage	Activities	Stakeholders (named where known)	TOTO's power
1. Raw materials	Kaolin, ball clay, feldspar, silica; brass/copper rod; resins; electronic components; high-purity Al ₂ O ₃ /AlN powders	Mining/minerals suppliers (domestic and imported); copper alloy mills; chemical makers; AlN/alumina powder makers (e.g. Tokuyama, Denka, Sumitomo Chemical — industry leaders, TOTO-specific not disclosed)	Moderate — commodity inputs; specialty powders more concentrated
2. Manufacturing	Casting, glazing, kiln firing (sanitary); faucet machining; Washlet assembly; ceramic sintering, electrode embedding, bonding (ESC)	TOTO plants: Kitakyushu/Kokura, Chigasaki, Shiga, Oita (Nakatsu – TOTO Fine Ceramics), Buzen (Fukuoka); Vietnam (4 plants incl. faucet plant Vinh Phuc), Thailand, Malaysia, India (Halol), China (Zhangzhou, Dalian; Beijing/Shanghai closed 2025), USA (Morrow & Lakewood, GA), Mexico	High — vertically integrated, proprietary process
3a. Distribution – housing	Wholesale, logistics, showroom, specification	Japan: plumbing/building wholesalers (e.g. Hashimoto Sogyo, Watanabe Pipe), home centres; US: plumbing distributors (e.g. Ferguson), showrooms, e-commerce; China/Asia: regional distributors	Strong in Japan (category leader); weaker overseas
3b. Distribution – ceramics	Direct to OEMs, co-design, qualification	Lam Research (primary); potentially Applied Materials, Tokyo Electron	Balanced — concentrated customer
4. Installers / integrators	Construction, remodelling, system-bath integration	Homebuilders (Sekisui House, Daiwa House, Sumitomo Forestry, Iida Group), general contractors, TOTO Remodel Club, TDY alliance (Daiken, YKK AP)	Moderate
5. End customers	Homeowners, hotels, offices, public facilities; memory and logic fabs	Households; hospitality chains; Samsung, SK hynix, Kioxia/SanDisk, Micron, (YMTC)	Brand pull in Japan; derived demand in chips
Competing suppliers at each stage		LIXIL, Panasonic, Takara Standard, Cleanup, Kohler, Roca, Geberit, Masco, V&B; ESC: AMAT/Lam in-house, Shinko (JIC), NGK, Kyocera, Niterra, Entegris, MiCo, CoorsTek	

Named distribution partners/customers are industry participants identified from public sources; TOTO does not disclose a supplier or customer list.

8. Financial Analysis: Five-Year Review by Division and Geography



Net sales (bars, JPY bn) and operating margin (line). FY3/27E–FY3/29E are our estimates.

8.1 Consolidated P&L (JPY bn, FY ending March)

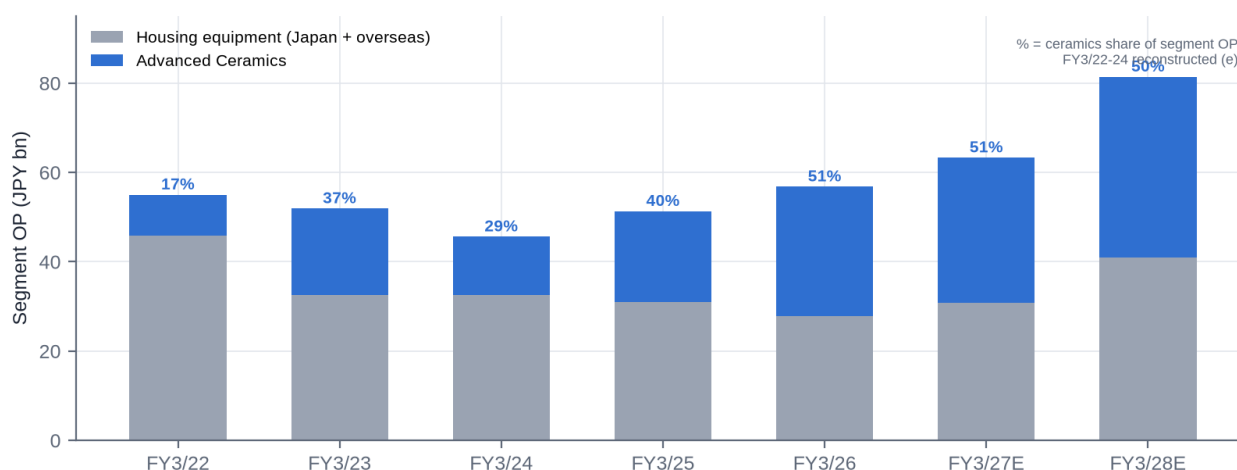
	FY3/22	FY3/23	FY3/24	FY3/25	FY3/26	5-yr CAGR / comment
Net sales	645.3	701.2	702.3	724.5	737.4	+3.4% (FY22–26)
growth	—	+8.7%	+0.2%	+3.2%	+1.8%	Price-led, volume flat
Gross profit	236.9	243.0	239.0	254.1	262.6	GM 36.7% → 35.6%
EBITDA	80e	79e	75e	83.5	88.1	D&A ¥34–35bn
EBITDA margin	12.4%e	11.3%e	10.7%e	11.5%	11.9%	
Operating income (EBIT)	52.2	49.1	42.8	48.5	53.8	Record in FY3/26
OP margin	8.1%	7.0%	6.1%	6.7%	7.3%	Trough FY3/24
Ordinary income	n/a	54.8	51.5	50.4	60.7	FX & dividend income
Net income (attrib.)	40.1	38.9	37.2	12.2	40.3	FY3/25: ¥34.1bn China impairment
EPS (¥)	~237e	~230e	~220e	~73e	243.0	
DPS (¥)	~90e	~100e	~100e	100	110	FY3/27 guide ¥120
ROE	~8.5%e	~8.0%e	~7.3%e	~2.3%e	7.7%	Target ≥12% by 2030

Source: TOTO kessan tanshin, results presentations (via secondary reporting). e = our reconstruction from reported growth rates/ratios; verify against EDINET before external use.

8.2 Segment / geographic sales and operating profit (JPY bn)

Segment	FY3/22e	FY3/23e	FY3/24e	FY3/25	FY3/26	Comment
Sales						
Japan Housing	~423	~451	~468	~481	479.7	Remodel ¥346.8bn (+1%); new-build ¥132.9bn (-3%)
China	~100	~100	~84	~67	53.8	-46% from peak; 2 plants closed
Americas	~45	~52	~59	70.5	75.6	CAGR ~14%; Georgia plant
Asia/Oceania	~38	~45	~48	~50	54.9	Taiwan, Vietnam, India
Advanced Ceramics	~36	~50	~42	50.3	67.4	+34% FY3/26; ¥85.5bn FY3/27 plan
Operating profit						
Japan Housing	~24	~15	~18	~21.8	20.3	Margin 4.2%; cost inflation > price
China	~14	~8	~4	~-4	-6.9	From best to worst region
Americas	~3	~3.5	~4.5	~5.1	~4.7	Tariffs, ramp-up costs
Asia/Oceania	~5	~6	~7	~8.2	10.2	Margin 18.6%
Advanced Ceramics	~9	19.4	~13	20.4	28.9	Margin 42.9%
Corporate/elim.	~-3	~-3	~-3	~-2.8	~-3.0	
Group OP	52.2	49.1	42.8	48.5	53.8	

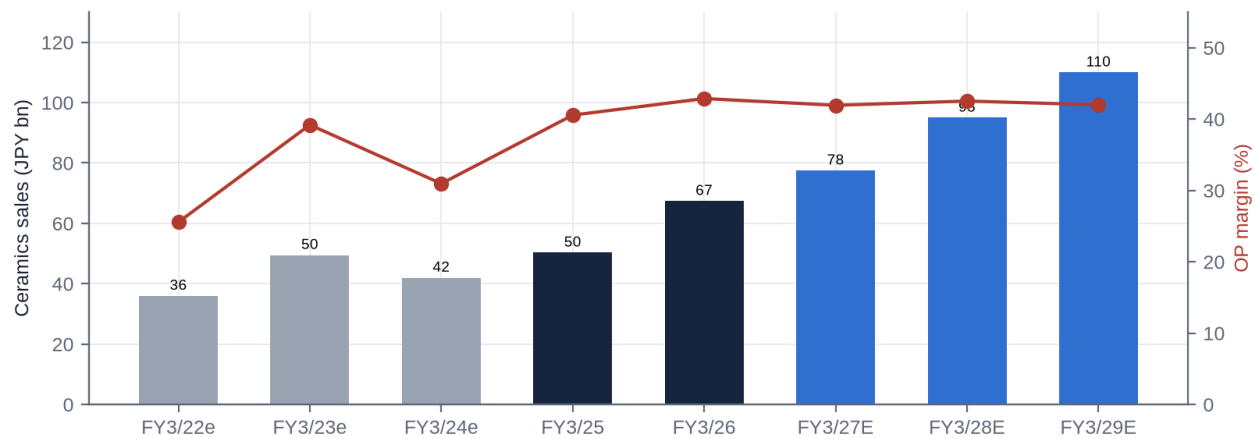
FY3/22–FY3/24 segment/regional splits are reconstructions (old three-segment basis: Japan Residential / Global Residential / New Domain) and are approximate; FY3/25 partly derived from FY3/26 growth rates. From FY3/26 TOTO reports Housing (Japan + overseas) and Ceramics.



Segment OP mix: housing vs Advanced Ceramics, with ceramics share of segment OP.

8.3 Trends and drivers

- **Revenue:** group sales grew only ~3.4% p.a. over five years and almost entirely through price. Japan volumes are flat to down (new-build structurally shrinking), China halved, Americas and Asia compounded at ~10–14%, and ceramics nearly doubled.
- **Margins:** OP margin fell from 8.1% to 6.1% (FY3/24) on China collapse, materials and energy inflation, and the FY3/24 memory downturn; it recovered to 7.3% purely on ceramics. Housing margin has fallen from ~11% to ~4.2%.
- **Net income:** FY3/25 was hit by the ¥34.1bn China impairment; FY3/26 included a further ~¥15bn China restructuring loss but benefited from FX/non-operating gains (ordinary income ¥60.7bn vs OP ¥53.8bn) and cross-holding sales.
- **EBITDA** ~¥88bn; D&A broadly flat at ~¥34–35bn but will rise with the ceramics and US capex programme.



Advanced Ceramics: sales (bars) and OP margin (line). FY3/22e–FY3/24e reconstructed; FY3/27E–FY3/29E our estimates.

8.4 Operational KPIs

KPI	Latest	Trend / comment
Japan remodel share of Japan housing sales	~72% (¥346.8bn)	Rising — structural; supports pricing
Japan housing starts (FY2025)	~711k (-12.9%)	Post-war low; building-code change pulled demand forward
Japan price increases	2023, 2024, 2025, Dec-2026 (8–13%)	Accelerating — cost pass-through test
Washlet cumulative shipments	>60m (Aug 2022)	Japan household penetration >80%
China sales / OP	¥53.8bn / -¥6.9bn	~40% of capacity closed; ~¥7bn improvement targeted
Americas sales	¥75.6bn (+7%)	Target ¥100bn by FY3/31; US local production >50%
Ceramics sales / OP margin	¥67.4bn / 42.9%	FY3/27 plan ¥85.5bn; Q1 +6% (timing)
ESC capacity	+20% from Jan 2027 (Buzen)	~¥30bn capex to FY2028; reported ~¥80bn 5-year
Capex intensity	~¥36.5bn PP&E (5% of sales)	Rising with ceramics + US
Overseas share of sales	~33%	WILL2030 target >40%
OP margin vs WILL2030 target	7.3% vs ≥12%	Gap closing only via ceramics

9. Earnings Calls, Sentiment and the Latest Result

9.1 What management has focused on — sentiment tracker

Event	Management focus	Sentiment (1–5)	Shift
Q2 FY3/24 (30 Oct 2023)	China property, guidance cut, cost control	2 – defensive	↓
Q2 FY3/25 (Oct 2024)	China loss, ¥34bn impairment, structural reform	1.5 – capitulation	↓
FY3/25 (28 Apr 2025)	China plant closures, ¥20bn buyback, US capacity	2.5 – reset	↑
Q1 FY3/26 (31 Jul 2025)	Weak Japan new-build, tariffs	2 – cautious	↓
Aug 2025 Americas briefing	¥100bn Americas target, US localisation	3 – constructive	↑
H1 FY3/26 (31 Oct 2025)	Ceramics surge, China reform near-complete; FY OP guide trimmed to ~¥50bn	3 – mixed	→
Q3 FY3/26 (30 Jan 2026)	Ceramics drives 9M OP +2%	3.5 – improving	↑
FY3/26 (30 Apr 2026)	Record OP, ceramics disclosure, capex, dividends, capital efficiency (post-Palliser)	4.5 – confident	↑↑
Q1 FY3/27 (31 Jul 2026)	ESC shipment timing, Middle East, Japan cost; full-year guide held; Dec price hike	3 – "on track" but defensive	↓

Our qualitative scoring based on disclosed results and press coverage of briefings; full transcripts not available to us.

Theme migration: 2023–24 was about China damage control; 2025 about restructuring and the Americas; from H2 2025 the narrative became ceramics/AI-NAND; after Palliser (Feb 2026) management adopted capital-efficiency language (segment disclosure, ROE, returns). The Q1 FY3/27 tone was steadier than the share price implied — guidance unchanged.

9.2 Latest result: Q1 FY3/27 (Apr–Jun 2026, reported 31 Jul 2026)

Item	Actual	y/y	Consensus	Variance
Net sales	¥168.6bn	+1.7%	~¥176.5bn	-4.5% miss
Operating income	¥8.07bn	-1.6%	~¥9.5bn (e)	~15% miss (e)
Ordinary income	¥9.37bn	+7.3%	n/a	FX gain ¥1.1bn

Item	Actual	y/y	Consensus	Variance
Net income	¥6.9bn	+9.5%	n/a	
EPS	¥42.0	+~11%	¥45.6	-8% miss
Ceramics sales / OP	~¥15.9bn / ~¥6.5bn	+6% / +10%		Shipment timing
Housing OP (incl. corporate)	~¥1.6–2.2bn	Japan -46%		Middle East ~¥3bn

- **Drivers:** ceramics grew but slower than the +27% full-year plan because of chunk shipment timing; Japan housing profit almost halved on procurement/labour costs and weak new-build; overseas housing OP -9% (Middle East order suspension, tariffs).
- **Margins:** group OPM 4.8% (Q1 seasonally weakest). Ceramics margin ~41%.
- **Guidance:** FY3/27 unchanged — sales ¥785bn, OP ¥60bn (+11.6%), NI ¥46bn, EPS ¥279.8, DPS ¥120. Q1 progress 13.5% of OP guide is below the ~18–20% normal pattern. Consensus (~¥62bn OP; EPS ~¥295) sits above guidance — downside risk to consensus, not necessarily to guidance.
- **Balance sheet flags:** none disclosed; watch ceramics inventory (shipment slip) and working capital ahead of the 1 Dec price rise (pre-buying in Q3, payback in Q4).
- **Market reaction:** -11.4% on 3 Aug. Combined with the ~¥9,500 June peak, this shows the stock was priced for uninterrupted ceramics acceleration; a modest timing slip removed ~¥140bn of value. That is a signal of momentum ownership — and of the opportunity if H1 confirms the slip was timing.

10. Earnings Forecasts: FY3/27E–FY3/29E

JPY bn	FY3/26A	FY3/27E	FY3/28E	FY3/29E	Key assumptions
Ceramics sales	67.4	77.5	95.0	110.0	+15% / +23% / +16%: NAND cryo upgrade; Buzen +20% capacity Jan-27
Ceramics OP (margin)	28.9 (42.9%)	32.5 (42%)	40.4 (42.5%)	46.2 (42%)	Pricing holds; depreciation step-up absorbed
Japan housing sales	479.7	485	495	500	Dec-26 price +8–13% offsets ~5–7% volume
Japan housing OP	20.3	17.0	22.0	23.0	FY27 hit by costs/Q1; FY28 full price benefit
China OP	-6.9	-1.5	+1.5	+2.5	~¥7bn restructuring benefit (partial)
Americas OP	~4.7	5.0	6.0	7.0	Tariff pass-through; Georgia utilisation
Asia/Oceania OP	10.2	11.0	12.0	13.0	Mid-single-digit growth
Corporate/other	~3.5	-3.8	-4.0	-4.2	
Group sales	737.4	765	800	833	Below ¥785bn guide in FY27
Group OP	53.8	60.0	78.0	87.5	In line with guide FY27
OP margin	7.3%	7.8%	9.8%	10.5%	
Non-operating, net	+6.9	+4.0	+3.0	+3.0	Dividends, interest; FX neutral
Extraordinary, net	~-9	+1.0	0	0	Cross-holding gains vs restructuring
Tax rate	~30%	29%	29%	29%	
Net income	40.3	45.7	57.0	63.8	
Avg shares (m)	165.7	163.5	162.0	161.0	Buybacks ~1% p.a.
EPS (¥)	243.0	280	352	396	Cons.: ¥295 / ¥353 / n/a
DPS (¥)	110	120	140	160	Payout ~40–45%

Our estimates. Financing costs negligible (net cash). Biggest swing factor: ceramics sales — each ±10% = ±¥3.5bn OP (±¥15/share).

- **Industry growth:** Japan housing equipment market flat to -1% real; Americas +5–7%; NAND etch equipment +20–30% p.a. 2026–28.
- **Market share:** stable in Japan; small gains in US; ceramics share held (risk: TEL cryo-etch).
- **Price/cost:** Dec-26 price rise worth ~¥25–35bn annualised gross; ~60% offset by cost inflation/volume.
- **Operating leverage:** ceramics incremental margin ~40%; housing incremental ~25% on price-led growth.
- **Dilution:** none (no converts/meaningful SBC); buybacks reduce share count ~1% p.a.

11. Capital Structure, Debt and Liquidity

Item (Mar 2026)	Value	Comment
Equity ratio	63.8%	Reported — very conservative
Total equity (e)	~¥530bn	Implied from ROE 7.67% and NI ¥40.3bn
Total assets (e)	~¥830bn	Implied from equity ratio
Cash & equivalents (e)	~¥110bn	Palliser cites ~¥76bn "excess cash"
Interest-bearing debt incl. leases (e)	~¥45bn	Mostly local borrowings/leases; no material bond maturities identified
Net cash (e)	~¥65bn	Net debt/EBITDA ≈ -0.7x

Item (Mar 2026)	Value	Comment
Investment securities / cross-holdings (e)	~¥60–76bn	~11% of net assets FY24; target <10% FY25, <5% by FY30
Interest cost	<¥1bn	Interest cover >50x
Credit rating	R&I: A+ area (to verify)	No covenants of note expected
Seasonal working capital	Moderate	Q3 build ahead of Dec price rise

Balance-sheet line items could not be retrieved directly from the FY3/26 filing in this research environment; (e) items are estimates derived from reported ratios and third-party commentary and should be verified. Conclusions (net cash, under-levered) are robust to the uncertainty.

Assessment: TOTO is materially under-levered. A move to an equity ratio of ~50% (still conservative) would free ~¥110bn of capacity; together with cross-holdings (~¥60–76bn) and excess cash, TOTO could return ~¥200bn (~20% of market cap) over 3–4 years without impairing its rating or ceramics capex. This is the core of the "round two" activist agenda.

12. Valuation: Peers, History and Regression

12.1 Peer comparison

Company	Ticker	Listed	Mkt cap (US\$bn)	EV/Sales	EV/EBITDA	EV/EBIT	P/E (fwd)	EBITDA mgn	EBIT mgn	Sales g	ND/EBITDA
TOTO	5332 JP	TSE	6.7	1.27x	10.6x	17.4x	21.8x	11.9%	7.3%	+1.8%	net cash
LIXIL	5938 JP	TSE	3.1	0.66x	9.0x	~26x	57x (t)	~7%	2.5%	+0.4%	~4x
Takara Standard	7981 JP	TSE	1.1	0.41x	n/a	5.4x	13.8x	n/a	7.5%	n/a	net cash
Cleanup	7955 JP	TSE	0.2	~0.15x	n/a	n/a	8.7x	n/a	2.9%	n/a	net cash
Rinnai	5947 JP	TSE	3.4	0.75x	5.3x	~7x	13.6x	14.1%	10.7%	n/a	net cash
Geberit	GEBN SW	SIX	23	6.9x	23.5x	~29x	29.8x	29.4%	~24%	+2.5%	~1.0x
Masco	MAS US	NYSE	13.4	2.1x	11.6x	~12.5x	14.6x	~19%	16.8%	-3%	~1.9x
Fortune Brands Innov.	FBIN US	NYSE	4.9	1.71x	16.2x	10.9x adj	12.7x	~19%	15.7% adj	-3.2%	2.9x
Villeroy & Boch	VIB3 GY	XETRA	0.6	0.47x	4.4x	~7x	n/m	~11%	6.8%	+1.8%	~1x
Arrow Home	001322 CH	SZSE	0.9	n/a	n/a	n/a	~136x (t)	n/a	~1%	-9.2%	n/a
Huida Sanitary	603385 CH	SSE	0.3	n/a	n/a	loss	n/m	n/a	-6.4%	-15%	n/a
NGK Insulators	5333 JP	TSE	11	~2.5x	12.0x	~17x	22.3x	~21%	14.2%	n/a	low
Kyocera	6971 JP	TSE	30–34	~1.1x	7.8x	~20x	16.7x	~13%	5.2%	+2.8%	net cash
Niterra	5334 JP	TSE	12.5	~2.3x	9.5x	~12x	~16x	~24%	18.9%	n/a	net cash
Ferrotec	6890 JP	TSE	~1.0	~1.15x	7.0x	~13x	9.1x	~16%	8.8%	+4%	~1x
MiCo	059090 KS	KOSDAQ	0.35	n/a	n/a	n/a	~16–18x	n/a	~10%	n/a	n/a
Kohler	—	Private	—	—	—	—	—	—	—	—	—
Roca (incl. Laufen)	—	Private	—	—	—	—	—	—	—	—	—
Duravit	—	Private	—	—	—	—	—	—	—	—	—
Jomoo / Hegii	—	Private	—	—	—	—	—	—	—	—	—
Panasonic Housing Solutions	(6752 JP parent)	Subsidiary	—	—	—	—	—	—	—	—	—
Shinko Electric (ESC)	(delisted 2025)	Private (JIC)	—	—	—	—	—	—	—	—	—
CoorsTek	—	Private	—	—	—	—	—	—	—	—	—

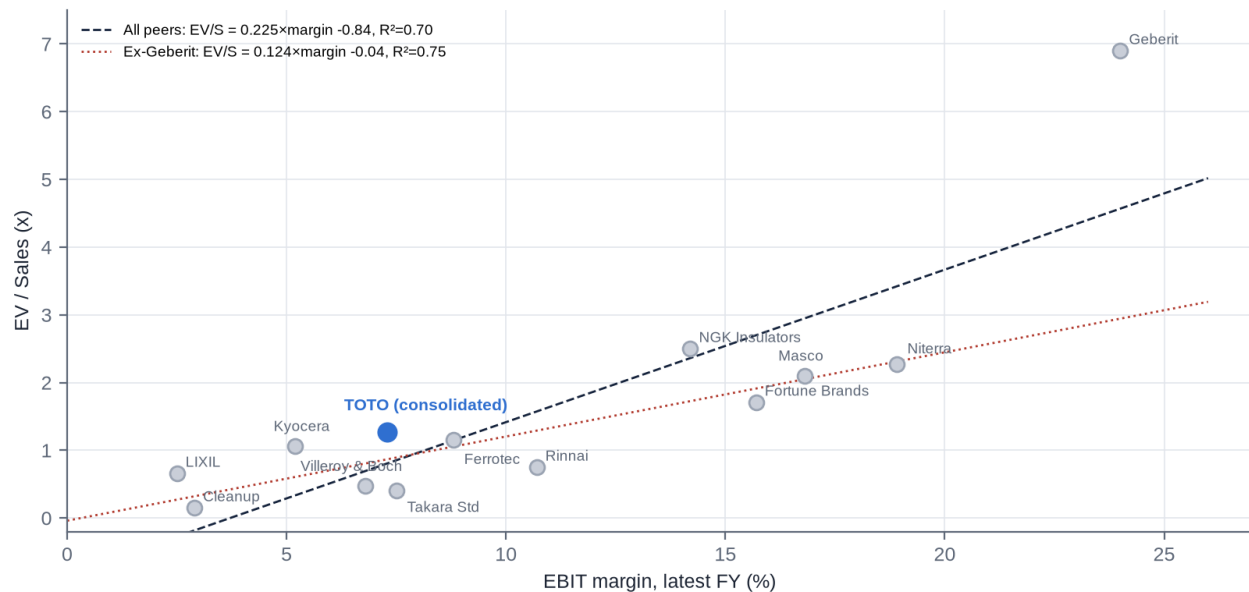
Peer data from company releases and data aggregators (Jul–Sep 2026; mixed dates); several EV/EBITDA and margin figures derived (–). TOTO EV/EBITDA on FY3/26A; P/E on our FY3/27E. Treat peer figures as indicative.

12.2 TOTO versus its own history

Metric	5-/10-yr low	5-/10-yr high	Median / average	Current	Read
P/E (trailing)	14.0x (Mar 2025)	54x (2025, impairment-distorted)	~28x (5-yr avg)	25.1x	Fair
P/E (forward)	~16x	~35x	~22x	21.8x	Fair
EV/EBITDA	6.1x (Dec 2024)	42x (10-yr)	13.7x (10-yr median)	10.6x	Cheap vs history
P/B	1.36x (Dec 2023)	~3.0x (2021)	~2.0x	~1.9–2.0x	Fair
Dividend yield	~1.3%	~2.9%	~2.0%	2.0%	Fair

Sources: companiesmarketcap, GuruFocus (TOTDY), YCharts; our calculations.

12.3 Regression: EV/Sales vs EBIT margin



EV/Sales vs latest-FY EBIT margin for 12 listed peers (building products and technical ceramics). Dashed = all peers; dotted = ex-Geberit.

Result. Across 12 listed peers $EV/Sales = 0.225 \times EBIT \text{ margin} - 0.84$ ($R^2 = 0.70$); excluding the Geberit outlier $EV/Sales = 0.124 \times \text{margin} - 0.04$ ($R^2 = 0.75$). At TOTO's consolidated 7.3% margin the regressions imply 0.81x–0.87x EV/Sales versus an actual 1.27x: on a consolidated basis TOTO screens ~45–55% expensive.

Why the consolidated read is misleading — the SOTP regression. Valuing the housing segment alone on the ex-Geberit line (4.2% margin $\rightarrow$ 0.48x sales $\rightarrow$ ~¥320bn EV) leaves ~¥615bn of EV for ceramics: ~9x ceramics sales, ~21x FY3/26 EBIT, ~19x FY3/27E and ~15x FY3/28E EBIT. For a 43%-margin, high-switching-cost consumable growing 15–25% p.a., that is reasonable against NGK (~17x EBIT, 14% margin) and Nittera (~12x, 19% margin). **Verdict: expensive as a housing company, fair-to-cheap as a SOTP.** The re-rating thesis therefore depends on investors continuing to capitalise ceramics separately — which better disclosure (Palliser's ask) supports.

12.4 Sum-of-the-parts (12-month target derivation)

Component	Metric	Multiple	Value (¥bn)	¥/share	Rationale
Advanced Ceramics	FY3/28E EBIT ¥40.4bn	18x EV/EBIT	727	4,460	Discount to NGK on concentration; premium on margin/growth
Housing equipment	FY3/28E EBIT ¥41.0bn	9x EV/EBIT	369	2,264	In line with Rinnai/Masco-ex-premium; China drag
Corporate costs	-¥4.0bn	9x	-36	-221	
Net cash (e)			65	399	
Cross-holdings, after tax (e)			45	276	
Equity value			1,170	~7,180	$\div$ ~163m shares
Cross-check: P/E	FY3/28E EPS ¥352	20.5x		7,216	Peer blend NGK 22x / Japan housing 14x
12-month target				¥7,200	+18% upside

13. M&A: Industry Deals, Takeover Potential and Takeout Multiples

13.1 Comparable transactions (2021–2026)

Date	Target	Acquirer	Value	EV/Sales	EV/EBITDA
Sep 2023 (closed early 2024)	Ideal Standard (ops)	Villeroy & Boch	~€600m	~0.81x	8.1x (5.5x post-synergy)
Jun 2023	Emtek/Schaub + Yale/August residential	Fortune Brands (from ASSA ABLOY)	~\$800m	~2.3x	n/a
Feb 2022	Elkay Manufacturing	Zurn (all-stock)	\$1.56bn	n/a	14.2x (9.8x post-synergy)
Late 2021	EZ-Flo International	Reliance Worldwide	\$325m	n/a	12x (~7x post-synergy)
Aug 2023	Bradley Corp.	Watts Water	\$303m	~1.5x	<8x post-synergy
Sep 2024	Kichler Lighting	Kingswood Capital (from Masco)	~\$125m	n/a	n/a (Masco paid ~\$550m in 2018)

Date	Target	Acquirer	Value	EV/Sales	EV/EBITDA
Dec 2023	KLAFS (saunas)	Kohler (from Egeria)	n/d		
May 2024	Kohler Energy (majority)	Platinum Equity	up to ~\$3bn		
2021	Royo bathroom furniture; Sanit	Roca	n/d		
Jan–Jul 2025	Fujitsu General	Paloma Rheem	~¥256bn	n/a	39% premium (6-m avg)
Dec 2023 – Jun 2025	Shinko Electric (ESCs, substrates)	JIC consortium (JIC/DNP/Mitsui Chem.)	~¥685–700bn	~3x (e)	~12–13x (e)
2022	CMC Materials	Entegris	~\$6.5bn	~5.4x	~17x (e)
2020	Permasteelisa	Atlas Holdings (from LIXIL)	undisclosed		

Sources: company releases and press (Börsen-Zeitung, Cleary Gottlieb, Watts 8-K, JIC). (e) = our estimate from reported financials.

13.2 TOTO's own acquisitions/investments (2016–2026)

Date	Target / project	Type	Cost	Multiple / outcome
Jul 2022	Vietnam 4th sanitary plant	Greenfield	n/d (~1m units/yr)	Successful — Asia OP +24% FY3/26
Mar 2024	Vinh Phuc (Vietnam) faucet plant	Greenfield	~¥10bn	Ramping
Nov 2024 / 2025	Morrow, Georgia (US) plant	Greenfield	US\$224m (~¥30bn)	+150% US capacity
Apr 2025	China restructuring (Beijing, Shanghai closures)	Exit	~¥15bn extraordinary loss (+¥34bn prior impairment)	Value-destructive legacy
Dec 2025 – Apr 2028	Buzen ESC kiln, Chigasaki R&D (~¥17bn), Nakatsu	Greenfield	~¥30bn (to FY2028); ~¥80bn 5-yr reported	High-return (segment ROIC well above WACC)
2016–2026	Acquisitions of companies	M&A	None material identified	TOTO is an organic grower

Pre-2016 consolidations (TOTO Washlet Techno 2007, Pagette 2009, TOTO Sanitechno 2011, TOTO Thailand 2013) for context.

13.3 Is TOTO a takeover target?

Pros	Impediments
Under-levered balance sheet (equity ratio 64%, net cash) — LBO-able housing cash flows	FEFTA: semiconductor parts are a "core" sector — foreign bidders face prior screening
Visible SOTP gap; ceramics asset scarce (Shinko taken private by JIC at ~12–13x EBITDA)	Stable shareholders: life insurers (~9%), employees, Japanese trust banks; Morimura-group culture
Japan M&A climate (METI guidelines, record take-privates, activism)	Size (~US\$7bn equity) and conglomerate structure — buyers want one half, not both
Activist presence (Palliser) can catalyse a strategic review	Management/brand heritage (founded 1917); no history of M&A; likely to resist hostile approach
Strategic logic for a ceramics carve-out to JIC/Japanese consortium or NGK/Kyocera	Lam Research dependency — buyer must underwrite customer change-of-control

Probability of whole-company takeover in 24 months: low (~5–10%). More plausible special-situation paths: (1) ceramics IPO/spin or minority sale (10–15%); (2) a large buyback/cross-holding unwind in STAGE3 (50%+). **Indicative takeout multiples:** whole company at 13–14x FY3/27E EBITDA (~¥1.25–1.35tn EV, ¥8,000–8,600/share, +30–40%); ceramics alone at 12–15x EBITDA / 18–22x EBIT (Shinko/CMC precedents and growth premium) = ¥600–800bn; housing at 8–10x EBITDA (V&B/Ideal Standard, Watts/Bradley).

14. Management Assessment

Executive	Role / tenure	Track record	Skin in the game
Shinya Tamura	President & Representative Director since Apr 2025 (age ~58)	Career TOTO; ran US and Vietnam subsidiaries (Vietnam capacity build-out successful); tasked with China rebuild; first year delivered record OP and adopted Palliser plan	Negligible personal stake (typical); pay ~¥172m incl. bonus (reported)
Noriaki Kiyota	Chairman since Apr 2025; President Apr 2020 – Mar 2025	Launched WILL2030 (2021); presided over China collapse and ¥34bn impairment but also ceramics rise and US investment	Negligible; ~¥139m pay (reported)
Tomoyuki Taguchi	Senior Managing Exec. Officer & CFO; Director	Led ¥20bn buyback (2025), segment disclosure upgrade (2026)	Negligible
Ryosuke Hayashi	Senior Managing Exec. Officer & CTO; Director	Oversees business divisions incl. Advanced Ceramics; ceramics OP 3x under watch	Negligible
Hiroshi Masumoto	Head of Ceramics Business Division	Ceramics capacity expansion; Lam supplier awards	n/d
Madoka Kitamura	Former Chairman (President 2014–2020); now senior adviser	Approved China expansion era	—

Sources: TOTO press releases (31 Jan 2025, 25 Feb 2026), annual securities reports via secondary sources. Pay figures reported, not verified against the yaho.

- **Archetype:** professional, lifetime-employee managers in a 109-year-old company; no founder or controlling family. Implication: continuity and risk aversion — good for the franchise, slow on capital efficiency. External pressure (Palliser, TSE) is what moved capital allocation.
- **Capital allocation:** mixed. *Good:* long-term commitment to ceramics R&D (since 1976) now paying off; Vietnam/US greenfields; first sizeable buyback (¥20bn, cancelled). *Poor:* China over-investment (Beijing/Shanghai plants) leading to ~¥49bn of impairment/restructuring; persistent over-capitalisation (equity ratio 64%, cross-holdings).
- **ROE trend:** ~8.5% (FY3/22e) → ~7.3% (FY3/24e) → ~2.3% (FY3/25e) → 7.7% (FY3/26) → ~8.6% guided; WILL2030 target ≥12%. ROIC on ceramics is far above WACC; housing ROIC ~3–4%, below WACC.
- **Red flags:** no related-party issues or accounting concerns identified. Compensation is moderate (bonus capped at ~0.6% of prior-year OP; little equity-linked pay — misaligned with share-price outcomes). Recurring product-safety recalls (Washlet fire risk series, faucets). Strategy is stable rather than pivot-prone; the 2026 "AI" framing came from an activist, not from promotional management.
- **Recent changes:** President change Apr 2025 (Kiyota → Tamura); 11 new executive officers and two R&D "Fellows" from Apr 2026; board cut from 13 to 10 at the June 2026 AGM (50% outside directors).

15. Board of Directors

Director	Position	Independent	Board tenure	Background / affiliations
Noriaki Kiyota	Chairman, Chair of the Board	No	Long-serving (President 2020–25)	Career TOTO
Shinya Tamura	President	No	Director since ≤2023	Career TOTO; ex-head of US and Vietnam subsidiaries
Tomoyuki Taguchi	Director, CFO	No	n/d	Career TOTO; finance, legal, HR
Ryosuke Hayashi or Naomiki Takeuchi	Director (inside)	No	n/d	Career TOTO (to confirm against 2026 AGM notice)
Junji Tsuda	Outside director	Yes	Several years	Former Chairman, Yaskawa Electric (to verify)
Kenji Yoshida	Outside director	Yes	New – Jun 2026	n/d
Masayuki Yoshioka	Director, full-time Audit & Supervisory Committee member	No	n/d	Career TOTO
Yukari Ienaga	Outside director, A&S Committee	Yes	n/d	n/d
Chiho Naganuma	Outside director, A&S Committee	Yes	n/d	n/d
Hidekazu Horikoshi	Outside director, A&S Committee	Yes	New – Jun 2026	n/d

Company with an Audit & Supervisory Committee. Post-23 Jun 2026 AGM: 10 directors, 5 independent. n/d = not determinable from sources available to us — complete from the 2026 Notice of Convocation before external use.

Board changes, 2021–2026

- Jun 2025: Naomiki Takeuchi joins (inside). Madoka Kitamura and Satoshi Shirakawa relinquish representative rights (Apr 2025) and step back to adviser roles.
- Jun 2026: Articles amended to reduce board size; board shrinks 13 → 10. Outside directors Shigenori Yamauchi and Yasushi Marumori retire; Kenji Yoshida and Hidekazu Horikoshi join. Independence rises to 50%. President re-elected with ~94% support (reported).
- 2021–2024 changes: routine rotation of inside directors; not fully reconstructable from available sources.

16. Shareholders, Activism and Short Interest

#	Shareholder	Type	% (Mar 2026, approx.)
1	Master Trust Bank of Japan (trust a/c)	Custodian – domestic passive/active funds	~17–19%
2	Custody Bank of Japan (trust a/c)	Custodian – domestic funds/pensions	~9.8%
3	Meiji Yasuda Life Insurance	Life insurer – stable/cross-holder	~6.1%
4	Nippon Life Insurance	Life insurer – stable holder	~3.2%
5	State Street Bank & Trust 505001	Foreign custodian (institutional)	~2.2%
6	TOTO Employee Stock Ownership Association	Insider / employees	~2.0%
7	State Street Bank & Trust 505103	Foreign custodian	~1.7%
8	BNYM as Agent / Clients Non-Treaty JASDEC	Foreign custodian	~1.5%
9	John Hancock Disciplined Value Int'l Fund	Foreign active fund	~1.4% (older list)
10	Other (JPM / Morgan Stanley custodial a/c)	Foreign custodian	~1%
—	Nomura AM group (5% report)	Institutional	8.0% (Jul 2025)
—	BlackRock Japan group (5% report)	Institutional	~4–6% (2024)
—	Palliser Capital	Activist hedge fund	"Top-20"; <5% (no large-holding report)

Sources: TOTO stock information page, half-year/annual securities reports, large-holding reports (via Kabutan). Directors' holdings are de minimis.

16.1 Activism: Palliser Capital (2026)

Palliser demand (Feb 2026)	TOTO response (30 Apr 2026)	Success
Disclose Advanced Ceramics properly (ESC for NAND cryo-etch, AD parts)	Two-segment reporting; ceramics sales/OP/margin disclosed; FY3/27 ceramics plan	Achieved
Invest more in Advanced Ceramics	~¥30bn capex to FY2028; Buzen +20% capacity; Chigasaki R&D; ~¥80bn 5-yr (reported Jun 2026)	Achieved
Restructure low-profit housing, especially China	China plant closures (pre-dated campaign); ~¥7bn improvement targeted	Partially (already under way)
Capital efficiency: appropriate equity ratio; review ~¥76bn excess cash and ~¥76bn cross-holdings	DPS ¥110 → ¥120; buyback (size unverified); cross-holding target <10% of net assets	Partial – the open item
Implied: unlock >55% upside (¥8,800+/share)	Stock peaked at ¥9,500 (Jun 2026); now ¥6,097	Achieved then reversed

Palliser publicly welcomed TOTO's "adoption of the Value Enhancement Plan" (7 May 2026). No shareholder proposals were filed at the June 2025 or June 2026 AGMs. Whether Palliser still holds the stake after the June peak is not disclosed; a renewed campaign around the STAGE3 plan (spring 2027) is plausible.

16.2 Short interest

- JPX short-position reports ($\geq 0.5\%$) have listed Barclays Capital Securities, Merrill Lynch International, Morgan Stanley MUFG, Nomura International, UBS AG, Integrated Core Strategies (Millennium) and Qube Research & Technologies at various dates — consistent with quant/market-maker and long-short pair trades (long ceramics peers/short TOTO or vice versa) rather than a fundamental short campaign.
- Margin long balance ~0.53m shares with a long/short ratio of ~5.1x (5 Jun 2026) — low relative to free float. We estimate total short interest at ~1–2% of shares; verify via JPX/karauri.net before trading.
- No published activist short report on TOTO identified.

17. Strategic Initiatives (2016–2026)

Initiative	When	What	Outcome / status
TOTO Vplan2017	to FY2017	Prior long-term plan: global growth, China/Asia expansion	China expansion later reversed
TOTO WILL2030	Apr 2021 –	Sales $\geq$ ¥1tn, OPM $\geq 12\%$, ROE/ROIC $\geq 12\%$, overseas >40% by FY2030	Off-track: sales ¥737bn, OPM 7.3%, ROE 7.7%
WILL2030 STAGE1	FY2021–23	Growth in China/Asia, ceramics, DX	Missed (China, memory downturn)
WILL2030 STAGE2	FY2024–26	¥175bn strategic capex; FY3/27 sales ¥785bn, OP ¥60bn	Ceramics ahead; housing behind; ends Mar 2027
China structural reform	2024–2025	¥34bn impairment; Beijing & Shanghai plants closed (~2,000 staff); consolidate Zhangzhou/Dalian	Execution done; profit recovery pending
Americas expansion	2024–2031	Georgia plant (US\$224m); ¥100bn sales by FY3/31; US production >50%	On track; tariff headwinds
Asia manufacturing hub	2014–2024	India (Halol), Vietnam 4th plant (2022), faucet plant (2024)	Successful — Asia OPM ~19%
Advanced Ceramics scale-up	2023–2028	Buzen kiln (+20% Jan 2027), Chigasaki R&D (Apr 2028), ~¥80bn 5-yr	Strong success; key value driver
Japan pricing discipline	2023–2026	Four consecutive price rises incl. 8–13% Dec 2026	Partial offset of cost inflation
Environmental building materials (Hydrotec)	1990s–	Photocatalytic tiles/coatings	Sub-scale; limited contribution
Morimura SOFC JV	2019 –	Solid-oxide fuel cells with NGK, Nitterra, Noritake	Small, pre-commercial
Capital-efficiency push	2025–2026	¥20bn buyback (cancelled), DPS growth, cross-holding reduction, segment disclosure	Under way — next step STAGE3 (Apr 2027)

18. Stakes in Other Companies

Holding	Type	Stake	Strategic relevance	Financial relevance
Noritake (5331 JP)	Listed – Morimura group	~1–3% (e)	Group heritage; ceramics know-how	Low
Nitterra (5334 JP)	Listed – Morimura group	~1% (e)	Group heritage; ceramics peer	Low–moderate (mark-to-market)
NGK Insulators (5333 JP)	Listed – Morimura group	small (unconfirmed)	Group heritage; ESC competitor	Low
Banks / insurers / distributors / homebuilders	Listed cross-holdings	various	Relationship holdings	Total ~¥60–76bn (e); being reduced
Morimura SOFC Technology	Unlisted JV	minority	Fuel-cell R&D	Negligible
TDY alliance (Daiken, YKK AP)	Commercial alliance	No equity	Remodel channel	n/a

Specified-investment details to be confirmed from the FY3/26 yuho ("proceeds to be redeployed.

"). Policy: cross-holdings <10% of net assets (FY2025), <5% by FY2030;

Relevance: strategically minor; financially, the portfolio (~6–8% of market cap) is a source of cash for buybacks — the most likely near-term "self-help" catalyst.

19. Shareholder Returns: Dividends, Buybacks and Treasury Shares

Fiscal year	DPS (¥)	Payout	Buyback	Cancellation	Comment
FY3/22	~90 (e)	~38%	—	—	
FY3/23	~100 (e)	~43%	—	—	
FY3/24	~100 (e)	~45%	—	—	Held flat
FY3/25	100	~137% (impairment year)	—	—	Maintained despite NI -67%
FY3/26	110	45.3% (DOE 3.47%)	Up to ¥20bn / 8m shares (4.7%), announced 28 Apr 2025; completed Aug 2025	Cancelled (2025)	First large buyback
FY3/27E	120 (¥60 + ¥60)	~43%	Announced Apr 2026 (size unverified; one source: up to 7.5m shares / ¥13bn)	n/d	Policy: payout ≥40%, progressive

Treasury shares: issued ~169.6m vs ~164.4m outstanding implies ~5m held (e). (e) items to be verified against TOTO's shareholder-return page and tanshin.

20. Adversarial Deep Dive: Bull, Bear, Pre-Mortem, Contrarian View

20.1 Bull case

- **Moat sustainability:** ESCs are process-of-record parts; requalification takes 12–24 months and risks yield. TOTO's 35+ years with Lam create a co-design loop; Palliser cites a ~5-year lead in cryo-ESC.
- **Growth levers:** NAND layer race (cryo-etch is the only economic path beyond ~300 layers), logic AD-coated parts (1nm-era R&D), replacement annuity, capacity +20% (Jan 2027) then more; US Washlet; China break-even; Dec 2026 price rise.
- **Earnings surprise potential:** FY3/28E EPS could reach ¥380–400 if ceramics hits ¥100bn+ and the price rise sticks (~+10% vs consensus).
- **Capital allocation:** ¥200bn of returnable capital (excess equity, cross-holdings, cash) could fund ~20% of market cap in buybacks over 3–4 years.

20.2 Bear case

- **1. Memory capex downturn (permanent-impairment risk: medium).** FY3/24 showed ceramics OP can fall ~35%; at peak margins, a 2027–28 NAND digestion would cut group OP by ~20% and the multiple simultaneously.
- **2. Customer/technology displacement.** Lam is the dominant customer; TEL's cryogenic etch, Lam dual-sourcing (NGK, Kyocera, in-house) or a shift to non-cryo architectures would permanently impair the segment's economics.
- **3. Housing structural decline.** Japan starts falling toward ~600k, labour shortages and Chinese competitors overseas could keep housing margins at ~3–4% indefinitely — and the housing business is still ~90% of capital employed.
- **High expectations?** Consensus EPS (~¥295 FY3/27) is above guidance (¥280) after a Q1 miss; the share price at ¥9,500 discounted ~30x FY3/28E. At ¥6,097 expectations are lower but not washed out (21.8x FY3/27E).

20.3 Pre-mortem: "It is October 2027 and TOTO is ¥3,800 — what happened?"

- H1 FY3/27 confirmed ceramics stagnation, not timing; Lam guided NAND customers to pause after 2026 pull-forward.
- The Dec 2026 price rise triggered share loss to LIXIL/Panasonic and a Q4 volume air-pocket; Japan housing OP fell below ¥15bn.
- China break-even slipped again; another impairment was taken.
- STAGE3 (Apr 2027) focused on capex and ¥1tn sales, with no ROE or buyback target; Palliser exited; the stock de-rated to a housing multiple (~14x P/E on ~¥270 EPS).

20.4 Are multiples too high?

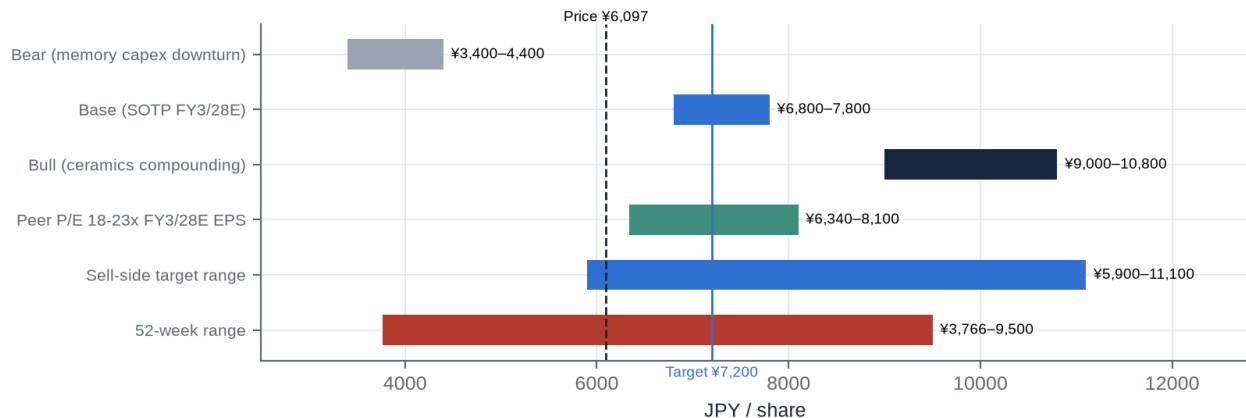
On consolidated metrics, yes (regression: ~50% premium to margin-implied EV/Sales; 21.8x FY3/27E P/E vs ~14x for Japanese housing peers). On SOTP, no: housing at peer multiples leaves ceramics at ~15x FY3/28E EBIT. The question is therefore not "is TOTO expensive" but "is the ceramics growth durable" — a cyclical-semiconductor question being underwritten by investors who mostly own TOTO as a defensive housing name.

20.5 Contrarian view: what the market refuses to see

The market is debating ceramics growth; it is ignoring the housing optionality and the balance sheet. Housing earned ~¥31bn of OP in FY3/25 even with China losing money; China at break-even plus the largest Japan price increase in a decade could add ~¥12–15bn of OP by FY3/28 — worth ~¥700/share at 9x — and that is not cyclical semiconductor exposure. Meanwhile ~¥200bn of excess capital is a call option on Japanese governance reform. Even if ceramics stalls, the downside is cushioned by self-help that the momentum-driven sell-off in August ignored.

21. Scenario Analysis

Scenario (prob.)	Ceramics sales FY3/29E	Ceramics margin	Housing OP FY3/29E	Group EPS FY3/29E	Multiple	Value / share	vs price
Bull (25%)	¥125bn (+23% CAGR)	44%	¥50bn	~¥470	Ceramics 22x EBIT; housing 12x	~¥10,000	+64%
Base (50%)	¥110bn (+18% CAGR)	42%	¥41bn (FY28)	~¥396	SOTP: 18x / 9x (FY3/28E)	~¥7,200	+18%
Bear (25%)	¥60bn (flat, downturn)	35%	¥26bn	~¥220	Ceramics 12x; housing 8x; P/E ~16x	~¥3,500	-43%
Probability-weighted						~¥6,975	+14%



Valuation range ("football field"), JPY per share.

Positioning implication: the asymmetry (+64% / -43%) is acceptable but not compelling at full size; we would size as a mid-weight event-driven position into H1 results (~30 Oct 2026) and add on confirmation of ceramics re-acceleration or a STAGE3 capital-return framework.

22. Questions for the CEO / Chairman (ordered by information value)

#	Question
1	What share of Advanced Ceramics revenue comes from Lam Research, and do you hold sole-source or dual-source positions on Lam's current cryogenic etch platforms?
2	Was the Q1 FY3/27 ESC shortfall purely shipment timing — what is the H1 order intake and book-to-bill for ceramics?
3	How do you see Tokyo Electron's cryogenic etch entry affecting your position — are you qualified, or in qualification, with TEL?
4	What return-on-capital hurdle and payback do you apply to the ~¥80bn ceramics programme, and what demand visibility underwrites capacity beyond Buzen?
5	Will the next mid-term plan (STAGE3) include explicit ROE (≥10–12%), equity-ratio and total-shareholder-return targets?
6	What is your target equity ratio, and how much of the ~¥60–76bn cross-shareholding portfolio will be sold by FY3/29?
7	Would the board consider structural options for ceramics (separate listing, tracking disclosure, minority partner) if the SOTP gap persists?
8	What is your realistic early read on volume elasticity to the 1 Dec 2026 price rise, and will LIXIL and Panasonic follow?
9	What exactly must happen for China to break even in FY3/27, and what is the remaining capital at risk there?
10	How much of US Washlet/toilet supply is now tariff-exposed, and what is the margin target for the Americas once Georgia is at full utilisation?
11	How do you allocate corporate costs between housing and ceramics, and what is the standalone ROIC of each segment?
12	What is the replacement-cycle share of ceramics revenue and how resilient was it in the FY3/24 downturn?
13	How will executive pay be linked to ROE/TSR rather than operating profit?
14	Is the ¥1tn sales target for FY3/31 still the right North Star, or should the company target profit and capital efficiency instead?

#	Question
15	What is your engagement status with Palliser and other large shareholders after the June 2026 AGM?

23. Short-Seller / Forensic View

23.1 Dismantling the bull case

- **Structural break:** the profit pool now depends on one technology (cryo-ESC), one customer (Lam) and one end market (NAND). A shift in etch architecture, a Lam dual-source decision or TEL share gains would hit >50% of group OP — with no offset from a ~4%-margin housing business.
- **Concentration shift:** if Lam moved 25% of its cryo-ESC volume to a second source, we estimate ceramics OP would fall ~¥8–10bn (-30%) and the SOTP would lose ~¥1,000/share.
- **Weaker moat than bulls think:** AMAT and Lam both have in-house ESC capability; Shinko is now JIC-backed with deep pockets; NGK and Kyocera have scale in AIN. Margins of 43% invite competition and customer push-back on price.
- **Most underestimated competitor:** Tokyo Electron — not as an ESC maker, but as a cryo-etch tool maker whose success would re-allocate the installed base away from Lam's platform where TOTO is embedded.
- **Worst capital allocation:** China (two plants built in 1995/2001, ~¥49bn written off 2024–26); decades of over-capitalisation (equity ratio ~64%, ROE <8%).
- **Incentives:** bonuses tied to operating profit and sales growth (¥1tn target), not ROE/TSR — encourages capex over buybacks.
- **What must hold for ¥6,097:** ceramics compounding ~18% p.a. at >40% margins through FY3/29 and housing margins recovering to ~5–6%. **If growth disappoints 20–30%** (ceramics FY3/29E ~¥80–90bn, EPS ~¥300–320), fair value on a 17x P/E is ~¥5,100–5,400 (-12% to -16%); with a cyclical de-rating to 14x, ~¥4,300 (-30%).
- **Single permanent-impairment scenario:** NAND makers adopt a non-cryogenic high-aspect-ratio etch route (or TEL wins the next node) while Lam qualifies a second ESC source — plausibility ~10–15% over 3 years.

23.2 Red flags and accounting risks

Area	Observation	Risk level
Revenue recognition	Ceramics recognised on shipment; Q1 FY3/27 "timing" miss shows lumpiness — watch for channel/stock-building at the customer	Medium
Segment reporting	FY3/26 change from three to two segments merges Japan and overseas housing — reduces visibility of China losses; corporate-cost allocation to ceramics opaque	Medium
Impairments	¥34.1bn (FY3/25) + ~¥15bn (FY3/26) China charges: recurring "one-offs"; remaining China assets and the Americas plant are the next tests	Medium
Non-operating income	Ordinary income exceeded OP by ¥6.9bn in FY3/26 (FX gains, dividends) — NI quality lower than headline	Low–Medium
Leases	Showrooms/logistics leases (J-GAAP: limited on-balance recognition) — understate EV slightly	Low
Related parties	None material identified; Morimura group JV is small	Low
Contingencies / provisions	Product-safety recalls (Washlet 96Z series fire risk; faucet leaks) — warranty provisions could rise	Low–Medium
Pensions	Japanese DB obligations; discount-rate sensitivity	Low
Stock-based compensation	Minimal; pay mostly cash bonus — no dilution, but weak alignment	Low
Goodwill / intangibles	Minimal (organic grower) — positive	Low
Cash flow	FCF ~¥35bn (OCF ¥71bn – capex ¥36bn); capex rising to ¥45–55bn will compress FCF as dividends rise	Medium
Cross-holding gains	Gains on sale flatter extraordinary income during the unwind	Low

24. Timeline and Catalyst Calendar

24.1 Key past events

Date	Event
1917	Toyo Toki founded in Kokura (Kitakyushu) — Morimura group
1976	Fine-ceramics research begins
1980	Washlet launched
1988	Mass production of electrostatic chucks
c.1990	Co-development with Lam Research begins
1995 / 2001	TOTO Beijing / TOTO East China (Shanghai) established
2007	Renamed TOTO Ltd.
2014	India plant (Halol, Gujarat)
Apr 2021	TOTO WILL2030 long-term vision
Jul 2022	Vietnam 4th plant

Date	Event
Oct 2023	FY guidance cut (China)
Apr 2024	WILL2030 STAGE2 (¥175bn strategic capex)
Oct 2024	¥34bn China impairment; stock -12.5%
Apr 2025	Tamura becomes President; China plant closures; ¥20bn buyback
Fall 2025	Georgia (US) plant production
Jan 2026	Goldman upgrade; stock +11%
Feb 2026	Palliser Capital Value Enhancement Plan
Apr–May 2026	Record FY3/26; ceramics disclosure; limit-up
Jun 2026	ATH ¥9,500; board cut to 10
Jul–Aug 2026	Q1 miss (-11.4%); Dec 2026 price rise announced

24.2 Upcoming catalysts

Date	Event	Importance
Oct–Nov 2026	Japan pre-buying ahead of 1 Dec price increase	Medium
21 Oct 2026	Lam Research Sep-quarter results (NAND/cryo commentary)	High
~30 Oct 2026	TOTO H1 FY3/27 results — ceramics timing, guidance	Critical
~mid-Nov 2026	Applied Materials results; Japan housing starts (monthly)	Medium
1 Dec 2026	Japan price increase effective (8–13%)	High
Jan 2027	Buzen ESC kiln complete (+20% capacity)	High
~29 Jan 2027	TOTO Q3 FY3/27 results	High
~late Apr 2027	FY3/27 results + WILL2030 STAGE3 mid-term plan (capital allocation, ROE, returns)	Critical
Jun 2027	AGM — possible activist engagement	Medium
Ongoing	US tariff decisions; Palliser/large-holding filings; cross-holding sales	Medium
Apr 2028	Chigasaki ceramics R&D building complete	Low

25. Data Notes and Disclaimer

This report was prepared on 4 October 2026 using publicly reported information (company press releases, results summaries and presentations as reported by Nikkei, Bloomberg, Reuters, Japan Times, Kabutan, Data-Max, Reform Online, LIMO, BusinessWire and data aggregators). Direct access to TDnet, EDINET and the TOTO IR site was not available in the research environment, so several items — notably the five-year segment history (FY3/22–FY3/24), balance-sheet line items, board backgrounds, treasury shares and short interest — are reconstructions or estimates and are marked "e", "~" or "n/d". Forecasts, valuations, scenario probabilities and the regression are the author's own. Peer data are indicative and dated Jul–Sep 2026. Verify all marked figures against primary filings before relying on them. This document is for professional investors, is not investment advice and does not constitute an offer to buy or sell securities.